

Supplemental information

Multimodal immune cell phenotyping in GI

biopsies reveals microbiome-related

T cell modulations in human GvHD

Sebastian Jarosch, Jan Köhlen, Sakhila Ghimire, Erik Thiele Orberg, Monika Hammel, Doris Gaag, Matthias Evert, Klaus-Peter Janssen, Andreas Hiergeist, André Gessner, Daniela Weber, Elisabeth Meedt, Hendrik Poeck, Elvira D'Ippolito, Ernst Holler, and Dirk H. Busch

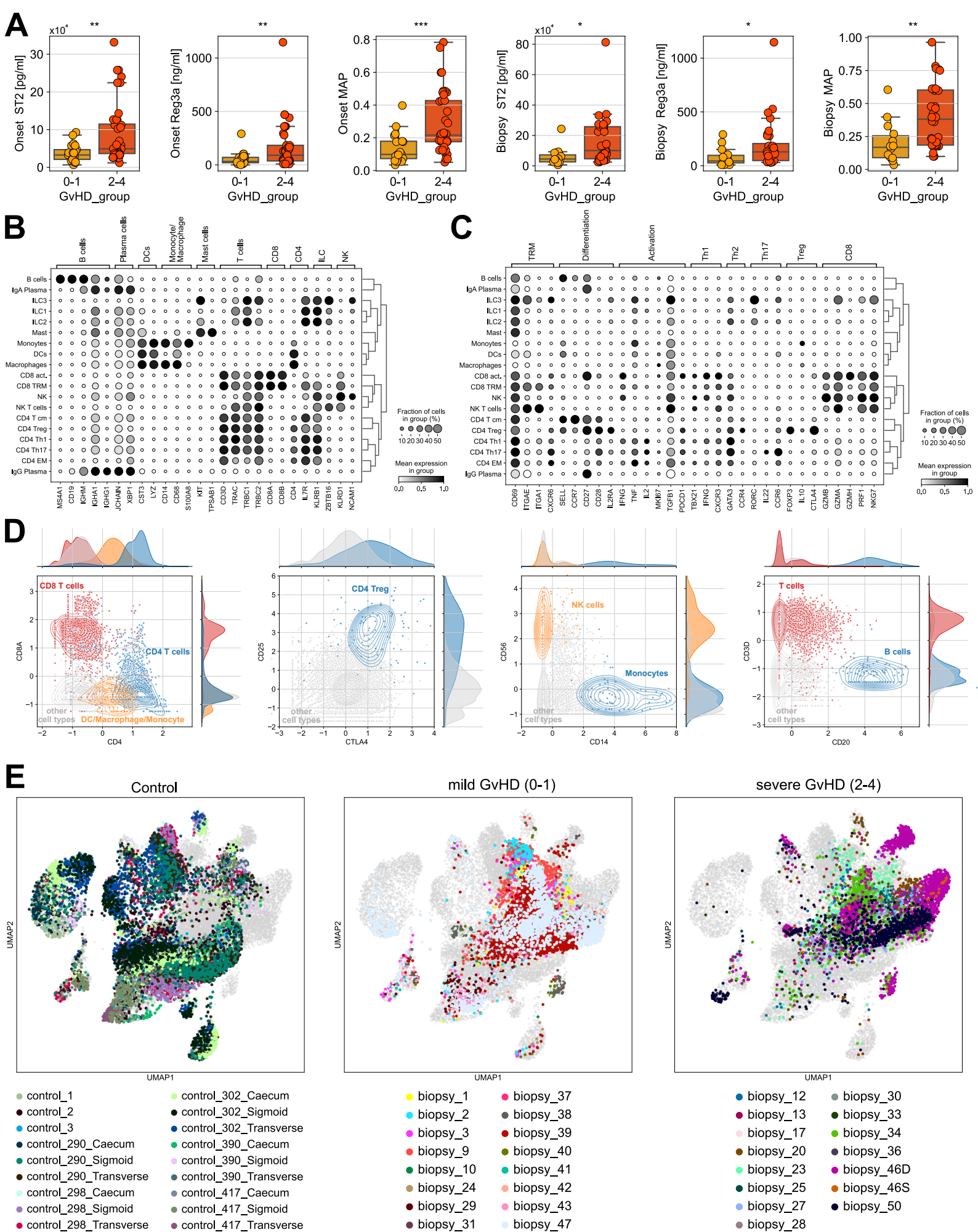


Fig. S1. Analyses of single-cell RNA sequencing data (related to Figure 1). (A) Clinical GvHD markers are plotted at the time of GvHD onset or biopsy sampling and grouped according to GvHD severity. (B-C) Expression of lineage (B) and phenotypic markers (C) for the annotated cell types. (D) Surface marker expression deduced from DNA-barcoded antibodies was used for validation of the annotated clusters based on transcriptomic data. (E) UMAP depicts the distribution of individual biosamples from control (left), mild GvHD (middle), and severe GvHD (right) sample groups. Statistical testing in (A) was conducted by Kruskal-Wallis H test.

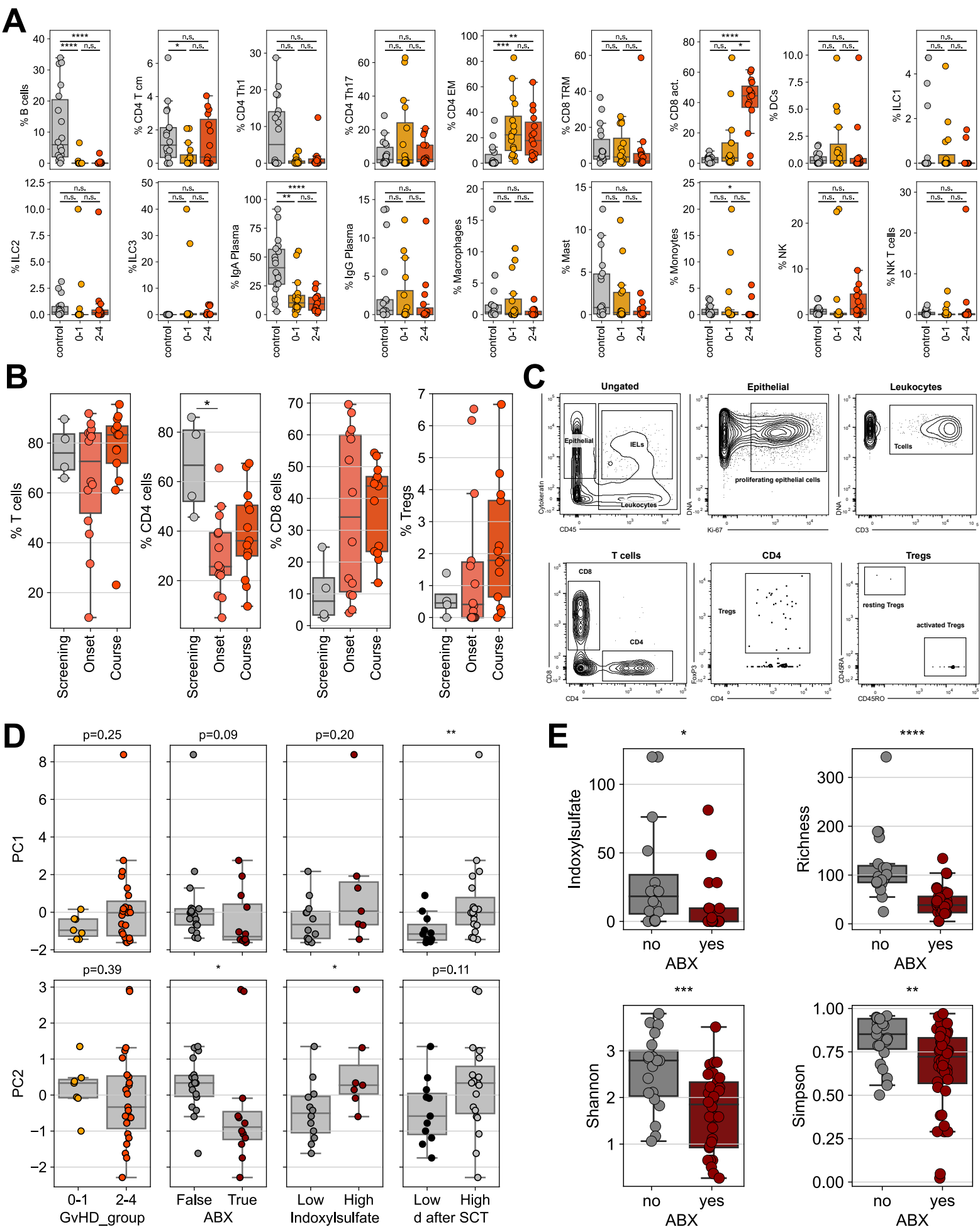


Fig. S2. Analyses of the gut immune infiltrate (related to Figure 2). (A) Cell type frequencies from the scRNA sequencing dataset, grouped into control, mild and severe GvHD samples. Each dot represents one biopsy. (B) Frequencies of gut-infiltrating T cell subsets. Each dot represents an individual biopsy. (C) Gating strategy for the quantification of ChipCytometry data. (D) Boxplots depict the significant variance explaining clinical features and the respective differences for the first (top) and the second (bottom) principal component. (E) Clinical microbiome parameters for the overall cohort (scRNA seq + ChipCytometry) at the time of the biopsy plotted regarding antibiotic treatment. Statistical testing in c) and d) was conducted by Kruskal-Wallis H test. Statistical testing in a) was conducted by one-way ANOVA followed by Dunn's multiple comparison with Holm-Bonferroni p-value adjustment.

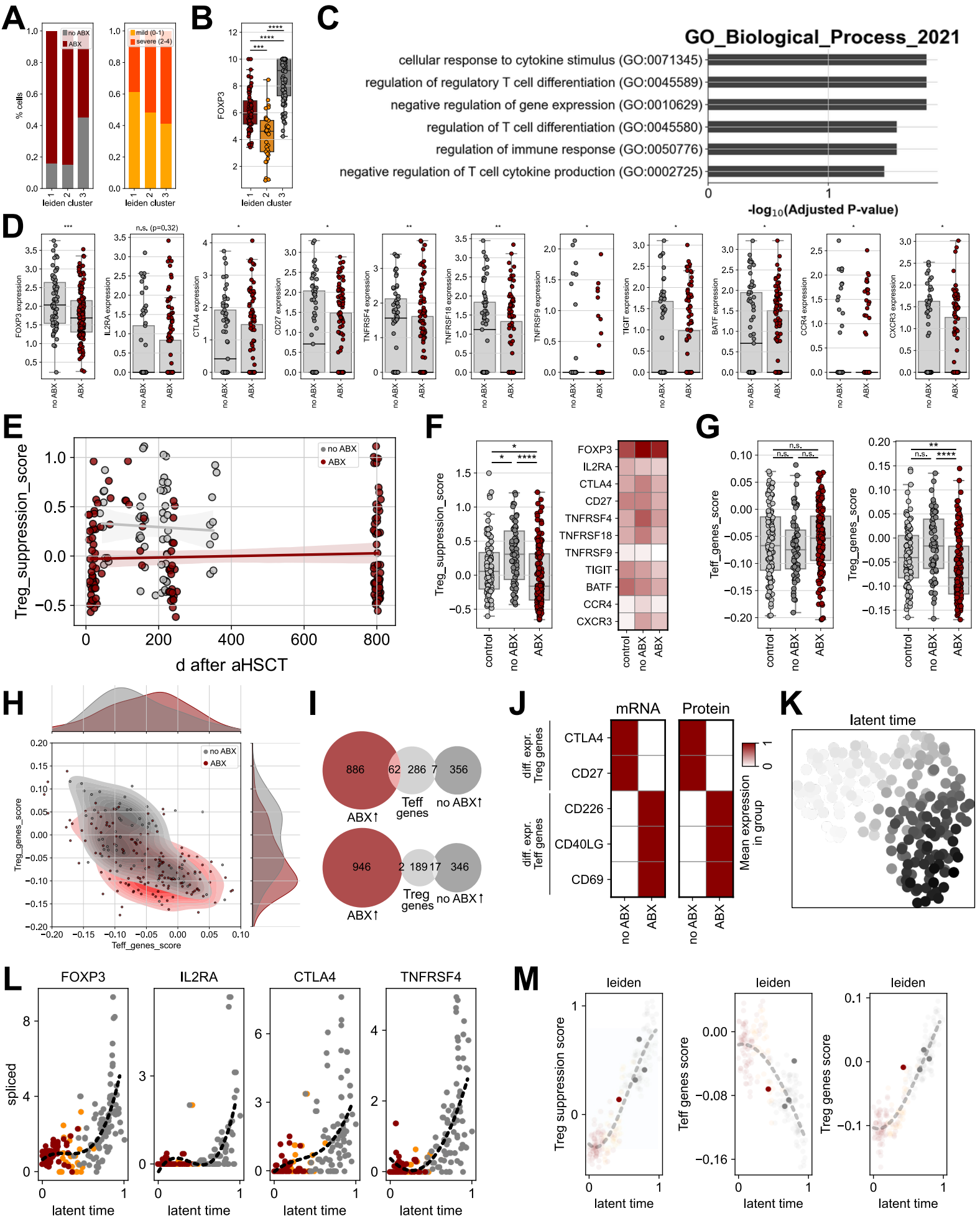


Fig. S3. Analyses of Treg transcriptional phenotypes in gut biopsies (related to Figure 3). (A) Distribution of cells from antibiotic-treated vs. untreated patients (left) and mild vs. severe GvHD patients (right) across the Leiden clusters. (B) FoxP3 expression in the individual clusters. Each dot represents one Treg. (C) Significant gene ontology terms for genes enriched in Tregs from non-antibiotic treated patients. (D) Expression differences of the individual genes representing the Treg suppression signature in Tregs from patients with or without antibiotic treatment. (E) Treg suppression score for antibiotic and non-antibiotic treated patients is plotted over time for each Treg of the dataset. (F) Boxplot showing the Treg suppression score for each Treg according to antibiotic treatment of aHST patients or control samples (left). Heat map indicating the average expression for all individual genes of the Treg suppression score (right). (G) Treg and Treg gene scores are depicted according to antibiotic treatment or control samples. (H) Treg vs. Treg signatures are plotted on a kernel density estimate (KDE) plot for each individual Treg. The color code resembles the antibiotic treatment. (I) Venn diagram depicting the gene overlap between ABX Treg-related genes, non-ABX Treg-related genes, and Treg genes (top) or Treg genes (bottom). (J) Overlap between transcriptional and surface protein signatures. (K) UMAP depicts the Treg latent time from early (light gray) to late (black). (L) Single cells are plotted according to the number of spliced transcripts of FOXP3, IL2RA, CTLA4, and TNFSF4 against the latent time. (M) Single cells from the same Treg clonotype are plotted according to their scores against the latent time. Statistical testing in (B), (F) and (G) was conducted by one-way ANOVA followed by Dunn's multiple comparison with Holm-Bonferroni p-value adjustment. Statistical testing in (D) was conducted by Kruskal-Wallis H test.

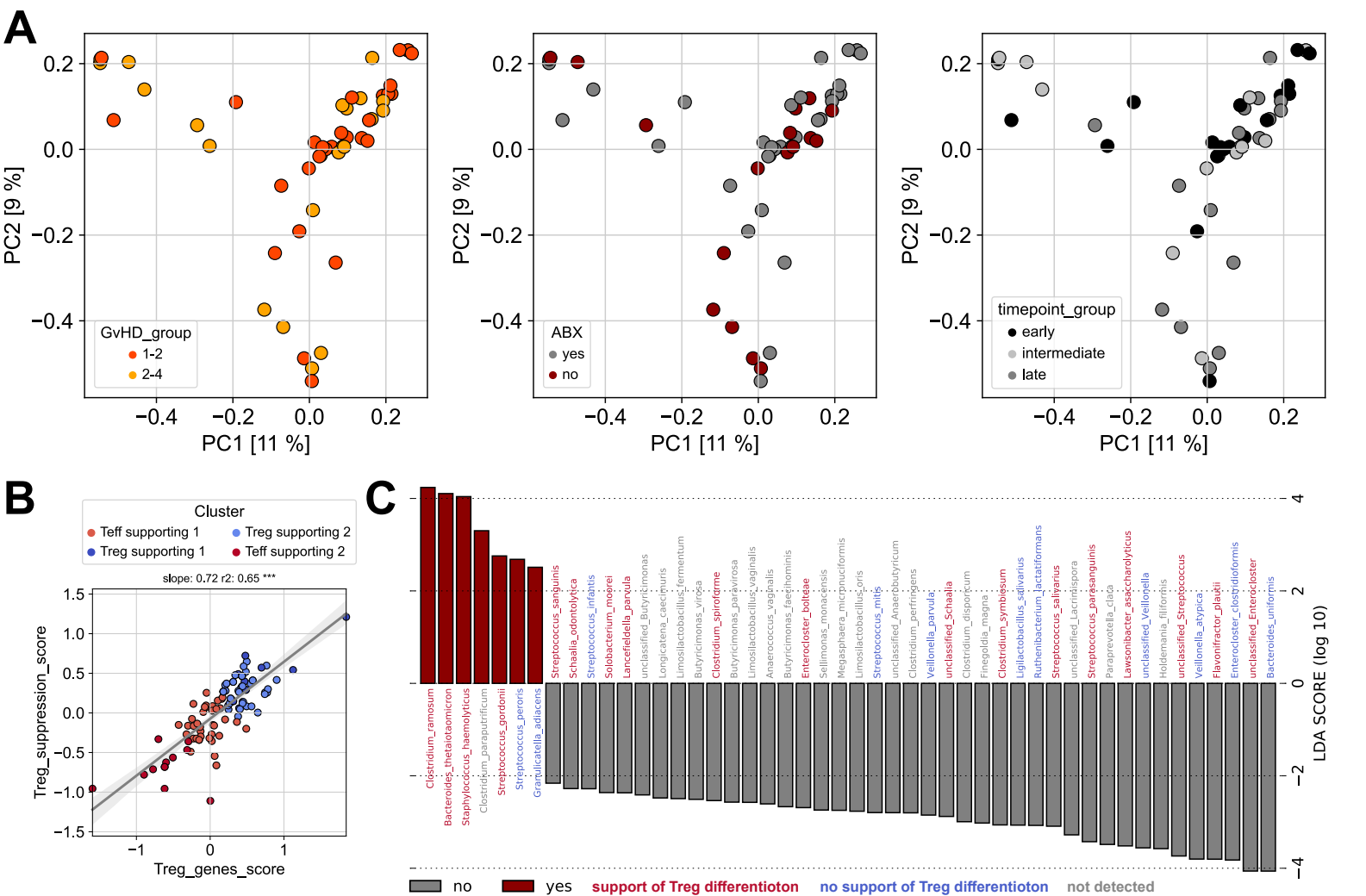


Fig. S4. Analyses of microbial taxa in relation to the phenotype of gut infiltrating Tregs (related to Figure 4). (A) PCoA analysis of the bacterial composition based on Bray-Curtis beta diversity measurement, color-coded according to GvHD severity (left), antibiotic treatment (middle), and the time point after HSCT (right). **(B)** Linear regression analysis between the influence of taxonomic features on the Treg gene score and Treg suppression score. Color coding refers to the k-Means clustering **(C)** LEfSe analysis of taxa abundance according to antibiotic treatment. Correlation analysis in (B) was done by Pearson's correlation.

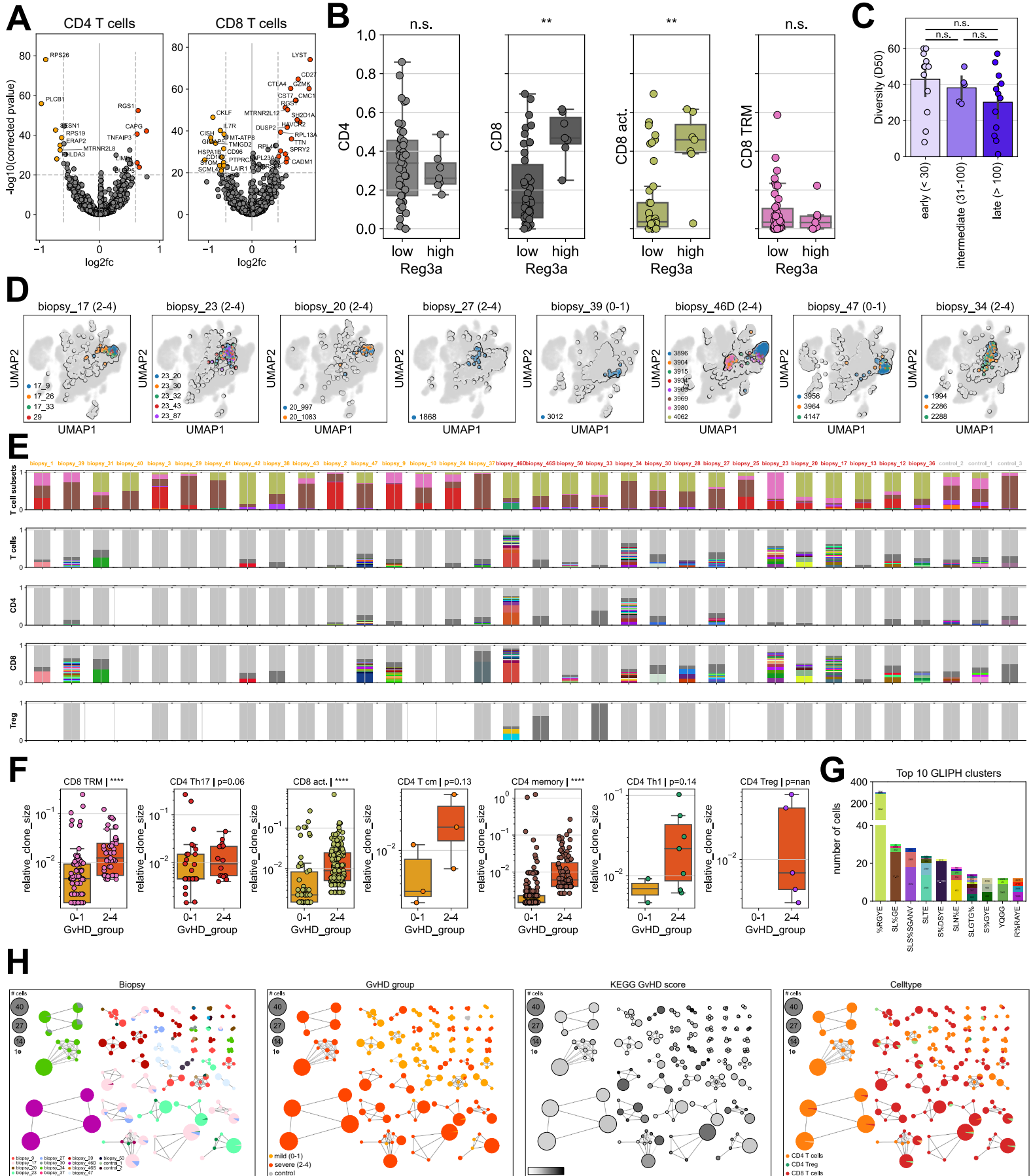


Fig. S5. Transcriptional and TCR clonotype features of gut T cells (related to Figure 5). (A) Volcano plot showing differential gene expression analysis according to GvHD severity for CD4 T cells (left) and CD8 T cells (right) **(B)** Frequency of T cell subsets grouped by the patient's Reg3a levels (higher or lower than the mean of all patients). **(C)** TCR repertoire diversity for all T cells of a biopsy, grouped according to the time point after aH SCT. **(D)** Clonotypes greater than 15 cells are depicted on the UMAP per biopsy. **(E)** Clonal expansion bar charts for each individual biopsy, grouped by GvHD severity (orange indicates mild GvHD; red indicates severe GvHD). **(F)** Relative clonal expansion grouped by GvHD severity for each T cell subset. **(G)** GLIPH analysis of CDR3-beta similarity showing clusters of clonotypes with similar CDR3 patterns. **(H)** Clonotype network analysis based on amino acid alignments of the CDR3ab regions. Statistical testing in (B) and (F) was conducted by Kruskal-Wallis H test. Statistical testing in (C) was conducted by one-way ANOVA followed by Dunn's multiple comparison with Holm-Bonferroni p-value adjustment.

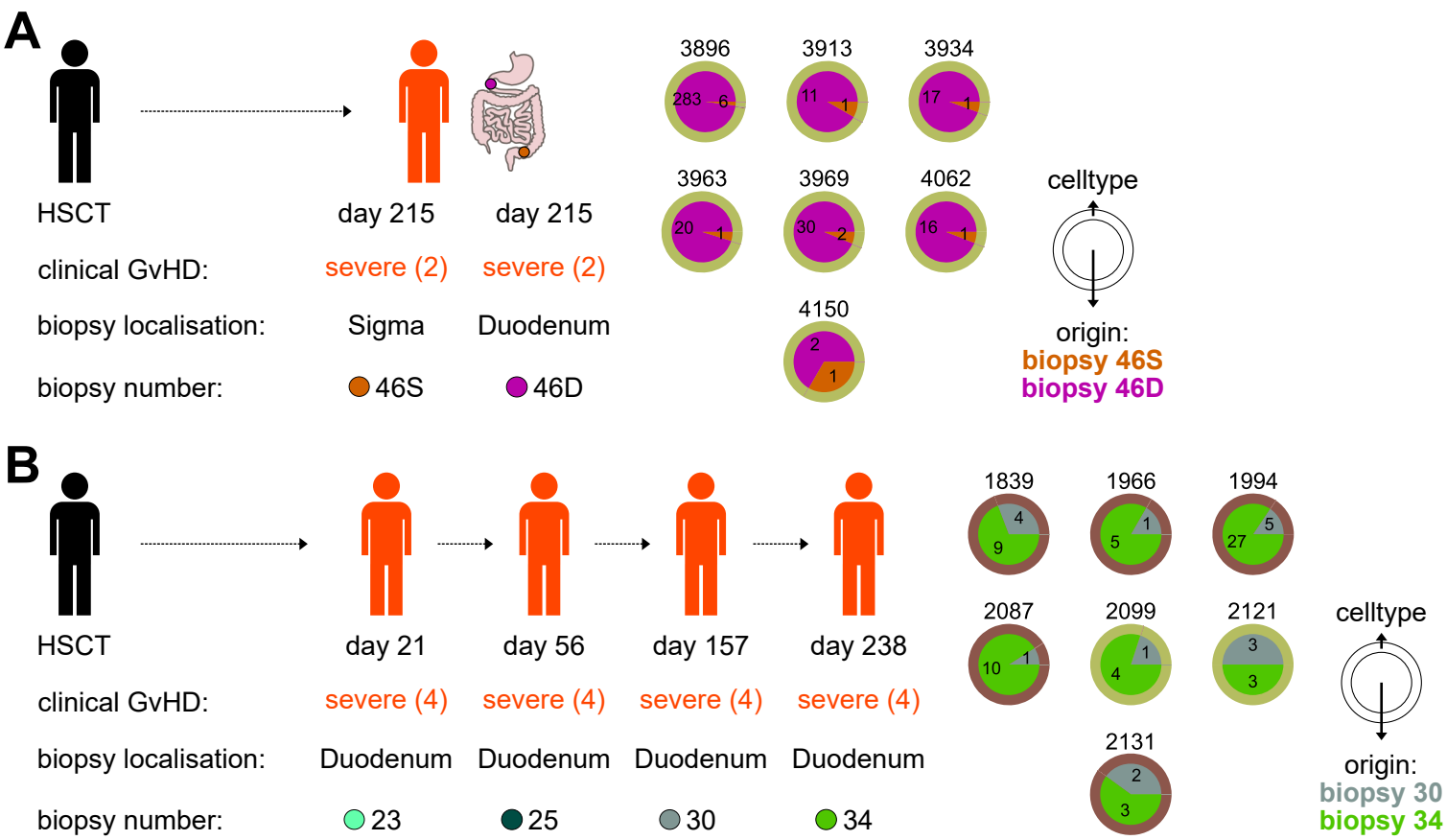


Fig. S6. Longitudinal analyses of TCR clonotypes (related to Figure 5). Biopsy time point and anatomical location (left), as well as the shared clonotypes along with their phenotype (right), are depicted for patient 54 (**A**) and patient 55 (**B**).

Patient	Dataset	Biopsy	GvHD (Lerner)	GvHD	d after aHCT	Origin	Type	clinical GvHD total	clinical GvHD GI	steroid treatment	steroid refractory	ABX	16S data	Indoxyl/sulfate [μmol/mol creatinine]	Richness	Simpson	Shannon	Biopsy ST2 [pg/ml]	Biopsy Reg3a [ng/ml]	Biopsy MAP	age of patient	gender	Diagnosis	Donor type	HLA	Donor age	Donor gender	Conditioning	Intensity	GvHD prophylaxis	Stem cell source	Bx Timepoint
Patient 01	ChpCytometry	biopsy_c01	1	1	15	Sigmoid	FFPE	4	4	yes	yes	yes	yes	81.220	103	0.3677527	2.7097513				55-50	m	B-NHL	unrelated	mismatch	25-30	m	Thio/Treo/Flud	RIC	CyAMTX	PBSC	Course
Patient 02	ChpCytometry	biopsy_c08	3	1	248	Sigmoid	FFPE	4	3	yes	yes	yes	yes	14.600	90	0.8646358	2.836934	79258	39	0.2362524	40-44	f	AML	unrelated	ident	55-60	m	FTM	RIC	CyAMTX	PBSC	Course
Patient 03	ChpCytometry	biopsy_c12	0	0	76	Sigmoid	FFPE	0	0	no	no	no	yes		97	0.9038373	2.6678665				45-50	m	MPN	unrelated	ident	45-50	m	FTM	RIC	CyAMTX	PBSC	Screening
Patient 04	ChpCytometry	biopsy_c13	0	0	39	Colon 60 cm	FFPE	2	1	yes	no	no	yes	1.190	70	0.5434547	1.1708946	85484	56	0.2691285	45-50	m	B-NHL	unrelated	ident	15-20	m	FTM	RIC	CyAMTX	PBSC	Screening
Patient 05	ChpCytometry	biopsy_c22	2	0	33	Colon 30 cm	FFPE	2	1	yes	yes	yes	yes	0.080	40	0.1455277	0.3644036	50197	29	0.1593457	20-24	m	AML	unrelated	ident	20-24	m	FTM	RIC	CyAMTX	PBSC	Screening
Patient 06	ChpCytometry	biopsy_c25	0	0	27	Sigmoid	FFPE	2	1	no	no	yes	yes	0.146	22	0.911352	2.7553655	26249	26	0.0956041	55-60	f	AML	unrelated	ident	40-44	m	FTM	RIC	CyAMTX	PBSC	Screening
Patient 07	ChpCytometry	biopsy_c26	1	1	21	Colon 20 cm	FFPE	2	1	yes	no	yes	yes	0.000	39	0.8248986	2.5412529	6521	34	0.0346206	55-60	m	MDS	haplo	haplo	30-34	m	FBM	RIC	PTCy/Tacro/MMF	BM	Onset
Patient 08	ChpCytometry	biopsy_c27	3	1	15	Sigmoid	FFPE	3	2	yes	yes	yes	yes	0.000	68	0.831985	2.0915994				65-70	f	MDS	haplo	haplo	65-70	f	FBM	RIC	PTCy/Tacro/MMF	BM	Onset
Patient 09	ChpCytometry	biopsy_c28	3	1	231	Sigmoid	FFPE	3	2	yes	yes	yes	yes	0.000	38	0.5131068	0.9496933	102515	294	0.422722	60-64	f	B-NHL	unrelated	ident	40-44	m	Treo/Flud	RIC	CyAMTX	PBSC	Onset
Patient 10	ChpCytometry	biopsy_c02	0	0	32	Colon 30 cm	FFPE	0	0	no	no	yes	yes	7.010	123	0.8873417	2.8714633				55-60	m	MDS/MPN	haplo	haplo	15-20	f	FBM	RIC	PTCy/Tacro/MMF	BM	Screening
Patient 11	ChpCytometry	biopsy_c03	3	0	15	Sigmoid 30 cm	FFPE	1	1	yes	yes	yes	yes	0.025	23	0.8518151	2.3284823	27107	8	0.0738706	45-50	f	T-NHL	unrelated	ident	35-40	m	FLANSAT/Vp16	Standard	CyAMTX	PBSC	Course
Patient 12	ChpCytometry	biopsy_c04	0	0	20	Colon 30 cm	FFPE	0	0	yes	no	yes	yes	48.500	35	0.7861175	1.9657442				55-60	m	B-NHL	sibling	ident	55-60	f	FTM	RIC	CyAMTX	PBSC	Screening
Patient 13	ChpCytometry	biopsy_c05	1	1	18	Duodenum	FFPE	0	0	no	no	yes	yes	0.000	56	0.6781073	1.8255593				50-54	m	MPN	unrelated	ident	50-54	m	FTM	RIC	CyAMTX	PBSC	Course
Patient 14	ChpCytometry	biopsy_c06	2	3	1	Colon 30 cm	FFPE	1	0	yes	no	yes	yes	101	101	0.8300362	2.4119171	92878	291	0.3973161	60-64	m	MDS	unrelated	mismatch	35-40	m	FLANSAT/Treo/Flud	RIC	CyAMTX	PBSC	Onset
Patient 15	ChpCytometry	biopsy_c07	1	1	34	Colon	FFPE	3	3	yes	yes	yes	yes	0.000	55	0.3834882	0.9216528				50-54	m	MPN	sibling	ident	60-65	m	FTM	RIC	CyAMTX	PBSC	Course
Patient 16	scRNAseq	biopsy_02	1	0	15	Sigmoid	SCS	0	0	yes	no	yes	yes	72	0.8388071	2.496326				45-50	m	B-NHL	sibling	ident	45-50	m	Thio/Treo/Flud	RIC	CyAMTX	PBSC	Screening	
Patient 17	scRNAseq	biopsy_10	1	1	188	Sigmoid	FFPE	2	1	yes	no	yes	yes	84	0.7707595	1.98206	243011	151	0.6046885	15-20	m	Hodgkin	unrelated	haplo	55-60	f	Thio/Treo/Flud	RIC	PTCy/Tacro/MMF	BM	Course	
Patient 18	scRNAseq	biopsy_03	0	0	17	Sigmoid	SCS	0	0	no	no	yes	yes	53	0.1364688	0.3890012				65-70	f	MDS	unrelated	mismatch	30-34	f	FBM	RIC	CyAMTX	PBSC	Screening	
Patient 19	scRNAseq+SP	biopsy_47	0	1	801	Duodenum	SCS	2	1	no	no	yes	no	2.750	39	0.8523522	2.2284087	331714	279	0.7506125	70-74	f	AML	unrelated	ident	55-60	m	FTM	RIC	CyAMTX	PBSC	Course
Patient 20	ChpCytometry	biopsy_c09	0	0	33	Colon 30 cm	FFPE	3	3	yes	yes	yes	yes	0.690	25	0.5385675	1.0631298				60-64	m	B-NHL	unrelated	mismatch	20-24	m	Treo/Flud	RIC	PTCy/Tacro/MMF	PBSC	Screening
Patient 21	ChpCytometry	biopsy_c10	0	0	62	Colon 40cm	FFPE	0	0	no	no	yes	yes	9.130	48	0.8689105	2.3257524	240514	1147	0.7834389	55-60	m	MDS/MPN	unrelated	mismatch	30-34	m	FBM	RIC	CyAMTX	PBSC	Onset
Patient 22	ChpCytometry	biopsy_c11	1	1	45	Sigmoid	FFPE	3	3	yes	yes	yes	yes								65-70	m	MPN	unrelated	mismatch	20-25	m	FBM	RIC	CyAMTX	PBSC	Onset
Patient 23	scRNAseq+SP	biopsy_31	1	1	357	Colon	SCS	2	1	yes	no	yes	no		35	0.590812	1.1109869	46917	8	0.1122666	55-60	f	B-NHL	unrelated	mismatch	40-44	m	TBI (8 Gy)/Flud	Standard	PTCy/Tacro/MMF	PBSC	Onset
Patient 24	scRNAseq	biopsy_24	1	1	23	Colon	SCS	2	1	no	no	yes	yes		27	0.7816501	1.9015282	62328	54	0.2143375	50-54	m	ALL	unrelated	mismatch	40-44	m	TBI (8 Gy)/Flud	Standard	CyAMTX	PBSC	Onset
Patient 25	scRNAseq+SP	biopsy_33	1	1	176	Colon	SCS	3	3	yes	yes	yes	yes	98	0.6575689	2.0980813	93979	59	0.23	55-60	m	AML	sibling	ident	55-60	f	ThioBu/Flu	RIC	CyAMTX	PBSC	Onset	
Patient 26	scRNAseq	biopsy_28	2	1	54	Colon	SCS	3	2	yes	yes	no	yes		90	0.9180458	2.7429322				50-54	m	MDS	sibling	ident	50-55	m	FTM	RIC	CyAMTX	PBSC	Course
Patient 27	scRNAseq+SP	biopsy_36	4	1	112	Sigmoid	SCS	3	3	yes	yes	yes	yes		143057	169	0.463	65-70	m	MDS	haplo	haplo	35-40	m	ThioBu/Flu	RIC	PTCy/Tacro/MMF	BM	Onset			
Patient 28	scRNAseq+SP	biopsy_34	0	0	64	Duodenum	SCS	2	1	yes	no	yes	yes		43004	80	0.179	55-60	m	AML	unrelated	mismatch	30-34	f	ThioBu/Flu	RIC	CyAMTX	PBSC	Onset			
Patient 29	scRNAseq+SP	biopsy_38	0	0	62	Duodenum	SCS	2	1	yes	no	yes	yes					55-60	m	AML	unrelated	mismatch	30-34	m	ThioBu/Flu	RIC	CyAMTX	PBSC	Course			
Patient 30	scRNAseq+SP	biopsy_39	0	1	20	Duodenum	SCS	2	1	no	no	yes	no					30-34	m	T-ALL	sibling	ident	25-30	m	TBI/Vp16	Standard	CyAMTX	PBSC	Course			
Patient 31	scRNAseq+SP	biopsy_37	0	0	22	Duodenum	SCS	0	0	no	no	yes	no		21081	35	0.087	60-64	m	AML	haplo	haplo	35-40	m	ThioBu/Flu	RIC	PTCy/Tacro/MMF	BM	Screening			
Patient 32	scRNAseq+SP	biopsy_42	1	1	28	Sigmoid	SCS	2	1	yes	no	yes	yes					60-54	m	MDS	haplo	haplo	25-30	m	FBM	RIC	PTCy/Tacro/MMF	BM	Onset			
Patient 33	scRNAseq+SP	biopsy_41	1	1	20	Sigmoid	SCS	2	1	yes	yes	yes	yes		47	0.3693404	1.042444				35-40	m	Hodgkin	unrelated	mismatch	40-44	m	TBI (12 Gy)/Flud	Standard	PTCy/Tacro/MMF	PBSC	Onset
Patient 34	scRNAseq+SP	biopsy_43	1	1	17	Sigmoid	SCS	2	1	yes	no	yes	yes					25-30	m	AML	sibling	ident	20-24	m	FBM	RIC	CyAMTX	PBSC	Onset			
Patient 35	scRNAseq+SP	biopsy_40	4	1	44	Sigmoid	SCS	3	2	yes	yes	yes	yes					60-64	m	AML	unrelated	mismatch	20-24	m	FBM	RIC	CyAMTX	PBSC	Onset			
Patient 36	ChpCytometry	biopsy_14	3	1	479	Colon	FFPE	3	3	yes	yes	yes	yes	11,900	189	0.9526639	3.8001475				50-54	m	ALL	unrelated	ident	15-20	f	TBI/Cy2	Standard	CyAMTX	PBSC	Course
Patient 37	ChpCytometry	biopsy_c15	2	1	327	Colon 50 cm	FFPE	4	3	yes	yes	yes	yes					30-34	m	AML	unrelated	ident	30-34	m	FBM	RIC	CyAMTX	PBSC	Course			
Patient 38	ChpCytometry	biopsy_c16	2	1	22	Sigmoid	FFPE	3	3	yes	yes	yes	yes					60-64	m	AML	unrelated	ident	30-34	m	FBM	RIC	CyAMTX	PBSC	Course			
Patient 39	ChpCytometry	biopsy_c17	3.5	1	64	Sigmoid	FFPE	3	2	yes	yes	yes	yes	51.5				50-54	f	CLL	unrelated	ident	45-50	m	FBM	RIC	CyAMTX	PBSC	Onset			
Patient 40	ChpCytometry	biopsy_c18	2	1	107	Colon	FFPE	3	3	yes	yes	yes	yes	0.000				30-34	m	MDS	unrelated	ident	30-34	m	Bu/Flu	RIC	CyAMTX	PBSC	Onset			
Patient 41	ChpCytometry	biopsy_c19	1	1	159	Colon 30 cm	FFPE	3	3	yes	yes	yes	yes	7,490	812907	69974	441	0.3610571	45-50	f	MDS	unrelated	ident	35-40	m	FBM	RIC	CyAMTX	PBSC	Onset		
Patient 42	ChpCytometry	biopsy_c20	2	1	150	Sigmoid	FFPE	1	0	no	no	yes	yes		95	0.8032723	2.1099094	75594	217	0.3289723	55-60	f	B-NHL	unrelated	ident	25-30	m	TBI (8 Gy)/Flud	Standard	CyAMTX	PBSC	Course
Patient 43	scRNAseq	biopsy_01	0	0	16	Sigmoid	SCS	0	0	yes	no	yes	yes		63	0.8007367	2.1436294				45-50	f	AML	unrelated	ident	30-34	m	FBM	RIC	CyAMTX	PBSC	Screening
Patient 44	ChpCytometry	biopsy_c21	1	1	180	Pars descendens	FFPE	4	4	yes	yes	yes	yes					55-60	m	AML	unrelated	ident	25-30	m	FTM	RIC	CyAMTX	PBSC	Course			
Patient 45	ChpCytometry	biopsy_c23	0	0	20	Sigmoid	FFPE	0	0	yes	no	yes	no					35-40	f	ALL	unrelated	mismatch	25-30	m	TBI/Cy2	Standard	CyAMTX	PBSC	Screening			
Patient 46	ChpCytometry	biopsy_c24	0	0	1320	Sigmoid	FFPE	0	0	no	no	yes	no					45-50	m	AML	sibling	ident	30-34	m	Meil/Treo/Flud	RIC	CyAMTX	PBSC	Screening			
Patient 47	scRNAseq+SP	biopsy_46D	0	0	215	Duodenum	SCS	3	2	no	yes	no	no					20-24	f	AA	unrelated	mismatch	30-34	m	TBI (2 Gy)/Cyto/Flud	RIC	CyAMTX	PBSC	Course			
Patient 48	scRNAseq+SP	biopsy_46S	2	1	216	Duodenum	SCS	3	2	no	yes	no	no					20-24	f	AA	unrelated	mismatch	30-34	m	TBI (2 Gy)/Cyto/Flud	RIC	CyAMTX	PBSC	Course			
Patient 49	scRNAseq	biopsy_23	1	1	21	Duodenum	SCS	4	4	yes	yes	yes	yes		24	0.7254844	2.0195581	33285	37	0.1243354	45-50	f	AML	unrelated	mismatch	15-20	f	ThioBu/Flu	RIC	CyAMTX	PBSC	Onset
Patient 50	scRNAseq	biopsy_25	1	1	56	Duodenum	SCS	4	4	yes	yes	yes	yes		101	0.9397428	3.1286887	80202	43	0.2433276	45-50	f	AML	unrelated	mismatch	15-20	f	ThioBu/Flu	RIC	CyAMTX	PBSC	Onset
Patient 51	scRNAseq+SP	biopsy_30	3	1	157	Duodenum	SCS	4	4	yes	yes	yes	yes		104	0.8959567	2.7569															

ID	donor	Origin	gender	age	derived from
control_1	1	colon	m	75-80	CRC resection
control_2	2	colon	m	80-84	CRC resection
control_3	3	colon	m	65-70	CRC resection
control_290_Caecum	290	Caecum	f	60-64	Gut cell atlas
control_290_Sigmoid	290	Sigmoid	f	60-64	Gut cell atlas
control_290_Transverse	290	Transverse	f	60-64	Gut cell atlas
control_298_Caecum	298	Caecum	m	55-60	Gut cell atlas
control_298_Sigmoid	298	Sigmoid	m	55-60	Gut cell atlas
control_298_Transverse	298	Transverse	m	55-60	Gut cell atlas
control_302_Caecum	302	Caecum	m	40-44	Gut cell atlas
control_302_Sigmoid	302	Sigmoid	m	40-44	Gut cell atlas
control_302_Transverse	302	Transverse	m	40-44	Gut cell atlas
control_390_Caecum	390	Caecum	f	65-70	Gut cell atlas
control_390_Sigmoid	390	Sigmoid	f	65-70	Gut cell atlas
control_390_Transverse	390	Transverse	f	65-70	Gut cell atlas
control_417_Caecum	417	Caecum	m	65-70	Gut cell atlas
control_417_Sigmoid	417	Sigmoid	m	65-70	Gut cell atlas
control_417_Transverse	417	Transverse	m	65-70	Gut cell atlas

Table S2 (related to Figure 1). Information on control samples.

Patient	Biopsy	B cells	CD4 T cells	CD4 Treg	CD8 T cells	Innate Immune cells	Plasma	Total cells	included	In T cell dataset	In Treg dataset
Patient 47	biopsy_01	0	37	0	20	1	23	81		yes	no
Patient 17	biopsy_10	0	13	1	9	1	13	36		yes	yes
Patient 59	biopsy_12	0	49	2	71	5	30	155		yes	yes
Patient 57	biopsy_13	0	40	4	102	0	27	169		yes	yes
Patient 56	biopsy_17	0	82	16	397	51	63	593		yes	yes
Patient 16	biopsy_02	0	155	7	5	5	31	196		yes	yes
Patient 58	biopsy_20	1	47	5	145	35	39	267		yes	yes
Patient 55	biopsy_23	0	128	6	297	6	75	506		yes	yes
Patient 26	biopsy_24	0	25	3	8	7	14	54		yes	yes
Patient 55	biopsy_25	0	16	1	4	5	16	41	yes	yes	
Patient 56	biopsy_27	0	284	9	221	13	45	563	yes	yes	
Patient 28	biopsy_28	0	34	1	36	4	8	82	yes	yes	
Patient 24	biopsy_29	0	88	2	9	3	123	223	yes	yes	
Patient 18	biopsy_03	1	78	2	17	36	12	144	yes	yes	
Patient 55	biopsy_30	0	54	4	52	1	4	111	yes	yes	
Patient 23	biopsy_31	0	16	0	25	0	7	48	yes	no	
Patient 27	biopsy_33	2	14	0	26	6	14	62	yes	no	
Patient 55	biopsy_34	0	460	18	145	49	43	697	yes	yes	
Patient 30	biopsy_36	0	38	4	90	12	6	146	yes	yes	
Patient 35	biopsy_37	0	215	1	9	6	20	250	yes	yes	
Patient 33	biopsy_38	0	10	4	14	76	4	104	yes	yes	
Patient 34	biopsy_39	0	1309	13	453	56	124	1942	yes	yes	
Patient 31	biopsy_40	0	2	0	2	36	0	40	yes	no	
Patient 37	biopsy_41	0	15	0	4	9	2	30	yes	no	
Patient 36	biopsy_42	0	6	2	32	5	3	46	yes	yes	
Patient 38	biopsy_43	0	21	1	9	16	49	95	yes	yes	
Patient 54	biopsy_46D	0	271	25	599	457	16	1343	yes	yes	
Patient 54	biopsy_46S	0	9	1	16	4	1	30	yes	yes	
Patient 19	biopsy_47	133	1102	60	461	71	259	2026	yes	yes	
Patient 39	biopsy_50	3	238	7	339	93	20	693	yes	yes	
Patient 60	biopsy_09	0	353	6	142	6	38	539	yes	yes	
Average		4,5	168,0	6,6	121,3	34,7	36,4	364,9	included total	31	26
SEM		4,3	54,1	2,1	29,8	14,8	9,3	93,4		31	31
control_1		267	92	24	91	58	302	834	included total	yes	no
control_2		63	112	22	74	29	479	779		yes	no
control_290_Caecum		161	353	56	86	54	237	947		yes	no
control_290_Sigmoid		56	231	26	37	61	418	829		yes	no
control_290_Transverse		8	305	21	49	97	405	885		yes	no
control_298_Caecum		9	378	24	58	11	411	891		yes	no
control_298_Sigmoid		23	215	18	69	46	487	858		yes	no
control_298_Transverse		11	309	27	60	56	427	890		yes	no
control_3		12	52	1	5	0	164	234		yes	no
control_302_Caecum		289	93	7	44	220	241	894		yes	no
control_302_Sigmoid		102	125	5	5	141	384	762		yes	no
control_302_Transverse		232	318	31	16	119	206	922		yes	no
control_390_Caecum		9	80	4	53	16	58	220		yes	no
control_390_Sigmoid		38	5	4	50	12	3	112		yes	no
control_390_Transverse		41	36	16	67	13	17	190		yes	no
control_417_Caecum		0	0	0	4	0	22	26		yes	no
control_417_Sigmoid		8	4	0	58	65	181	316		yes	no
control_417_Transverse		2	0	0	0	7	98	107		yes	no
Average		73,9	150,4	15,9	45,9	55,8	252,2	594,2	included total	18	0
SEM		22,7	31,6	3,5	6,8	13,6	39,6	83,4		18	18

Table S3 (related to Figure 1). Information on cell numbers per sample in the scRNA seq cohort.

clinical parameter	description	p value PC1	PC1 log2fc	p value PC2	PC2 log2fc
GvHD_group	mild vs. severe GvHD	0,249198756	-1,165354126	0,390840537	0,15966819
clinical GvHD GI	gastrointestinal GvHD (high/low according to mean)	0,100458026	-1,580035508	0,902435841	-0,08635076
d after aHSCT	time after transplantation (early/late according to mean)	0,008127113	-1,646333837	0,106556149	-0,45909635
Lifetime (month)	lifetime after transplantation	0,724873454	0,479385067	1	0,10068215
status	alive/dead	0,742695932	-0,404084655	0,453034014	0,12888868
steroid treatment	steroid treatment (yes/no)	0,678302466	-0,715656513	0,640765863	-0,22519878
steroid refractär	refractory to steroid treatment (yes/no)	0,241017389	-1,208842166	0,605940351	0,07118119
ABX	antibiotic exposure	0,090391715	0,716400358	0,011085166	0,4488193
Indoxylsulfate	Indoxylsulfate [μ mol/mmol creatinine] (high/low according to mean)	0,204893894	-1,293560509	0,042522478	-0,94294378
Richness	16S richness (high/low according to mean)	0,496979396	-0,065506882	0,341629673	-0,22900347
Shannon	16S Shannon diversity (high/low according to mean)	0,47719662	-0,663502498	0,858954923	0,15949243
Bakterien16S	16S qPCR (high/low according to mean)	0,617075077	-0,409194053	0,317310508	-0,83089121
Pilz28S	28S qPCR (high/low according to mean)	0,617075077	-0,409194053	0,317310508	-0,83089121
CCgrpXIVa	Frequency of Clostridia Cluster XIVa taxa (high/low according to mean)	0,654720846	-1,17271749	0,654720846	-0,57612408
EC copies	E. coli copies qPCR (high/low according to mean)	0,563702862	0,00811338	1	0,25905722
BCoAT	Butyryl-CoA-transferase qPCR (high/low according to mean)	0,654720846	-0,089917231	0,179712495	1,87940276
Biopsie ST2	Biopsie ST2 [pg/ml] (high/low according to mean)	0,853100389	-0,573893812	0,13852341	0,84216848
Biopsie Reg3a	Biopsie Reg3a [ng/ml] (high/low according to mean)	0,55411313	0,120402595	0,932646639	-0,02341679
Biopsie MAP	Biopsy MAP (high/low according to mean)	0,934191641	-0,194780793	0,679708154	-0,21246908
age of patient	(high/low according to mean)	0,582090388	0,141186894	0,611452987	-0,25315491
gender	male vs. Female	0,714506867	-0,063246297	0,175210944	-0,35050483

Table S4 (related to Figure 2). Principal component dependency for all clinical parameters.

Treg suppression	Treg genes			Teff genes			
FOXP3	ABHD11	HLA-DRB1	SWAP70	ACSL6	EHD4	NBEA	SHF
IL2RA	ACTA2	HLA-DRB5	TCF19	ACVR1B	ELMO2	NBL1	SIPA1L2
CTLA4	ADAM28	HOXB2	TCHP	ACVR1C	EPHA1	NCOA7	SKIL
CD27	ADAT2	HOXB3	TEP1	ADAMTS10	EPHA4	NELL2	SLC22A17
TNFRSF4	ADTRP	HPGD	TIAM1	ADAMTS17	EPS8	NEO1	SLC29A2
TNFRSF18	AKAP5	HS3ST3B1	TIGIT	ADCY9	ETS2	NFIA	SLC30A4
TNFRSF9	AKIP1	HSPA1L	TIMD4	ADGRE1	EXD2	NFKB1	SLC36A4
TIGIT	AKR1C3	IKZF2	TNFRSF13B	ADGRE5	FADS3	NFKBIA	SLC39A8
BATF	ARAP3	IKZF4	TNFRSF9	AFF3	FAM89A	NINJ1	SLC40A1
CCR4	ARHGAP11A	IL1R1	TNIP3	AGAP1	FBP1	NLRP3	SMAD5
CXCR3	ARHGAP25	IL1R2	TOP2A	AGAP3	FBXL8	NLRP6	SMIM3
	ARHGEF12	IL2RA	TOR4A	AIF1	FHL1	NME4	SNTB1
	ASF1B	IL32	TOX	ALOX5	FKBP2	NOL3	SORBS3
	ASPM	INPP1	TOX2	ALPK1	FKBP5	NR3C2	SOS2
	ATP1B1	INPP5F	TPP1	AMIGO1	FLOT1	NR4A1	SPATA6
	ATP6V0A1	IRAK3	TPX2	AMPD2	FOS	NR4A2	SPEF2
	BARD1	IRF4	TRIB1	ANK3	FOSB	NT5C3B	SPEG
	BFSP2	ITGB8	TSHR	ANKRD42	FZD3	OSBPL5	SPRY1
	BLM	JSRP1	TTC22	ANXA1	GADD45G	OSM	SREBF1
	BLZF1	KHDC1	TTN	APBA2	GAS2L1	OTUD7A	SSBP2
	BUB1	KIAA1841	TYMS	APOL3	GAS7	P3H3	STARD10
	C16orf74	KIF18A	UBALD1	APP	GCLM	PCSK5	STOM
	CAMK1	KLHL2	VANGL1	AREG	GCSAM	PDE4D	STXBP1
	CARD16	LAYN	VAV3	ARHGEF40	GIMAP7	PDZD4	SUSD4
	CARD17	LIMD1	XYLB	ARL4A	GIMAP8	PELI2	SVIL
	CASP1	LMCD1	YPEL2	ARMCX2	GLUL	PFN2	TAF4B
	CASP8	LSP1	ZBP1	ATF3	GNAO1	PHLDB2	TANC1
	CBR4	MACC1	ZBTB32	AZIN2	NGN7	PIK3C2B	TBC1D16
	CCDC141	MARCH3		B3GNT9	GPR160	PIM3	TC2N
	CCM2	METTL7A		BACH2	GPR35	PLAC8	TCEA3
	CCRL2	MFGE8		BAIAP3	GRASP	PLAG1	TCF7
	CD27	MGST2		BAMBI	GSAP	PLAUR	THEMIS
	CD79A	MPST		BCL2A1	H1F0	PLEKHA7	TIMP1
	CD79B	MYB		BCL7A	HBEGF	PLEKHG5	TLE1
	CDCA2	MYL6B		BHLHE40	HID1	PLK2	TLR2
	CDCA7L	MYOF		BRSK1	HIPK2	PLXDC1	TMEM121
	CDHR3	NABP1		BTBD11	HOOK1	PLXND1	TMEM170B
	CDK14	NCAPG		C16orf54	HS6ST1	PM20D2	TMEM184B
	CDKN2A	NDC80		C1orf162	HSF4	PMEPA1	TMEM200A
	CEACAM4	NINJ2		C3orf52	ID1	PNP	TMEM71
	CENPE	NTHL1		CACFD1	ID2	POMGNT2	TMIGD2
	CENPF	NUF2		CAMK2N1	IER3	PPM1L	TMOD2
	CENPJ	NUSAP1		CASS4	IER5	PRKCA	TNFAIP3
	CEP128	OAS1		CBR3	IFI44L	PRKCE	TNFRSF10B
	CHST2	OGT		CCDC112	IGFBP3	PTGER4	TNFRSF13C
	CIITA	PAFAH2		CCDC184	IGFBP4	PTK2	TOP1MT
	CPNE2	PCED1B		CCND1	IL15	PTPRK	TPM2
	CSF2RB	PECR		CCND3	IL2	PWWP2B	TRABD2A
	CTLA4	PGM2L1		CCR7	IL41	RAB20	TRADD
	CTNNAL1	PIK3R3		CCSAP	IL4R	RAB27B	TRAT1
	CTTNBP2NL	PITPNM2		CD226	IL5RA	RALA	TRIP10
	CXCR6	PM20D1		CD300A	IL7R	RALGPS2	UBE2E2
	DCLRE1A	PMAIP1		CD40LG	INPP4B	RARA	ULBP2
	DCP1B	POLE2		CD69	INSR	RARG	UNC13B
	DDI2	PRDM8		CDC42EP4	IQSEC2	RASD1	USP44
	DUSP10	PRKAG1		CDH23	IRF2BP2	RASGEF1A	USP46
	DUSP4	PROB1		CDKN1A	ITGA1	RASGEF1B	USP6NL
	ENTPD1	PTGS1		CERS6	JUN	RASGRF2	UST
	EPSTI1	PYHIN1		CFH	KCNQ1	RASSF2	VEGFB
	ERI1	RAB31		CFL2	KCTD3	RASSF6	VIPR1
	ETV7	RAB37		CHD7	KDM6B	RBMS1	WDR86
	FAM174B	RACGAP1		CHN2	KIF19	RELB	WNT1
	FAM184A	RAD51AP1		CHPF	KLF4	RGL1	WNT10B
	FDXR	RASGRP4		CKAP4	LATS2	RGS16	XBP1
	FHL3	RGPD2		CLU	LMO4	RHOB	YES1
	FITM2	RTKN2		CMPK2	LMTK3	RHOU	ZBED3
	FOXP3	RTP4		CNN3	LONRF1	RIMKLB	ZBTB10
	GALNT3	RYR1		COL1A1	LPGAT1	RNASE6	ZBTB18
	GCNT1	SASS6		COL6A1	LPIN2	RND1	ZBTB42
	GEN1	SFT2D1		CTSL	LRP1	RNF122	ZFP36
	GINS2	SGMS1		CXCL16	LYPD3	RNF130	ZFP36L2
	GIPR	SHMT2		CYB561	LYSMD2	RNF144A	ZFYVE9
	GK	SHTN1		DDIT4	MAGI3	RNF207	ZNF165
	GLCE	SIRPG		DENND5A	MAML3	ROBO3	ZNF208
	GLRX	SLC12A6		DHRS3	MAN1C1	RRAGD	ZNF219
	GPR155	SLC1A4		DISC1	MAP9	RRAS2	ZNF358
	GSN	SLC2A8		DLG4	MAPRE3	RSAD2	ZNF439
	GSTA4	SLC36A1		DMXL2	MARVELD1	RUFY3	ZNF467
	GXYLT1	SLC43A1		DSE	MATK	S1PR1	ZNF502
	HDAC9	SLC46A3		DST	MBNL3	SAV1	ZNF629
	HELLS	SLC9A7		DTX1	MCOLN2	SBF2	ZNF703
	HES1	SMC6		DUSP1	MEST	SCML1	ZNF814
	HIBCH	SMTN		DUSP2	METTL1	SCML4	ZNF827
	HIST4H4	SOX4		DUSP5	MID1IP1	SEC14L2	ZSCAN18
	HLA-DMA	SPATS2L		DUSP8	MID2	SECTM1	ZSWIM5
	HLA-DMB	SPON2		DYNC2H1	MMP15	SEMA4C	
	HLA-DOA	SRGAP3		DYNLT3	MPP7	SERPINB6	
	HLA-DPA1	ST8SIA4		ECHDC3	MTUS1	SFMBT2	
	HLA-DPB1	STAM		EDA	MYADM	SGK1	
	HLA-DRA	SUOX		EGR2	MYO6	SH2B3	

Table S5 (related to Figure 3). Gene lists for Treg suppression-, Treg-, and Teff- scores.
Bold genes refer to overlap between Treg suppression and Treg genes